GROUP 23 Numpcom 2 Classes

Class layouts (The currently implemented ones)

namespace numpcomp

BasicRootFinder
(no data members: stateless)
+ bisection(f, a, b, tol, max_iter)
+ newton_raphson(f, df, x0, tol, max_iter)

AdvancedRootFinder
(no data members: stateless)
+ secant_method(f, x0, x1, tol, max_iter)
+ fixed_point_iteration(g, x0, tol, max_iter)

Interpolator
(no data members: stateless)
+ linear_interpolate(x, x_vals, y_vals)
+ chebyshev_nodes(a, b, n)

BasicIntegrator
(no data members: stateless)
+ trapezoidal_rule(f, a, b, n)
+ midpoint_rule(f, a, b, n)

AdvancedIntegrator
(no data members: stateless)
+ simpsons_rule(f, a, b, n)
+ romberg_integration(f, a, b, max_iter)

MatrixSolver (for week 4)
(no data members: stateless)
+ gaussian_elimination(A, b)

BasicODESolver
(no data members: stateless)
+ euler(f, t0, y0, t_end, h)
+ heun(f, t0, y0, t_end, h)

AdvancedODESolver
(no data members: stateless)
+ rk4(f, t0, y0, t_end, h)
+ rk3(f, t0, y0, tend, h)

NB
- No arrows because no inheritance.
- All 8 classes inside namespace numcomp

PICTURE

You may delay, but time will not.